

Temperature Controlled DC Fan

Cad to Simscape



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ABSTRACT

This project focuses on designing and simulating a DC fan system controlled by temperature. The fan was first modeled in SolidWorks, where all parts like the propeller blades, housing, and shaft were created and assembled. After completing the CAD design, the model was exported and imported into MATLAB Simscape for dynamic simulation. During this process, some challenges appeared due to compatibility issues between SolidWorks and Simscape file formats. These issues were resolved by using intermediate XML files and the correct `smimport` command, ensuring smooth import of the model. The fan's rotation was controlled using a revolute joint with torque input which allowed realistic spinning motion.

To simulate real-world conditions, a control logic was developed that turns the fan ON when the temperature rises above 40 degrees and turns it OFF when the temperature cools down. This temperature threshold can be changed manually. Additionally, to ensure smooth starting and stopping of the fan, a ramp function was implemented in the control logic. The ramp had a slope of +2 and a start point at 1 which gradually increased the torque input to avoid sudden jerks during fan operation.

The simulation results demonstrated smooth transitions in the fan's motion in response to temperature changes, validating the control strategy and the effectiveness of the ramp function for smoothness. Overall, this project successfully combined CAD modeling, mechanical simulation, and control systems to create a realistic and functional temperature-controlled DC fan model. It highlights the importance of integrating design and simulation tools to analyze mechanical systems before physical prototyping.

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1. OBJECTIVES

The main objectives of this project are to design a complete DC fan assembly, including the housing and propeller, using SolidWorks and to successfully import the CAD model into MATLAB Simscape Multibody for dynamic simulation. The project aims to resolve software compatibility issues between SolidWorks and Simscape by utilizing intermediate data files. Additionally, the project focuses on implementing a smooth startup of the fan through ramp-based torque input and developing an effective temperature-based ON/OFF control logic to automate the fan operation. Finally, the simulation results will be analyzed to study the torque and dynamic behavior of the fan system under varying conditions

The Model for this Project is shown in Figure 1.

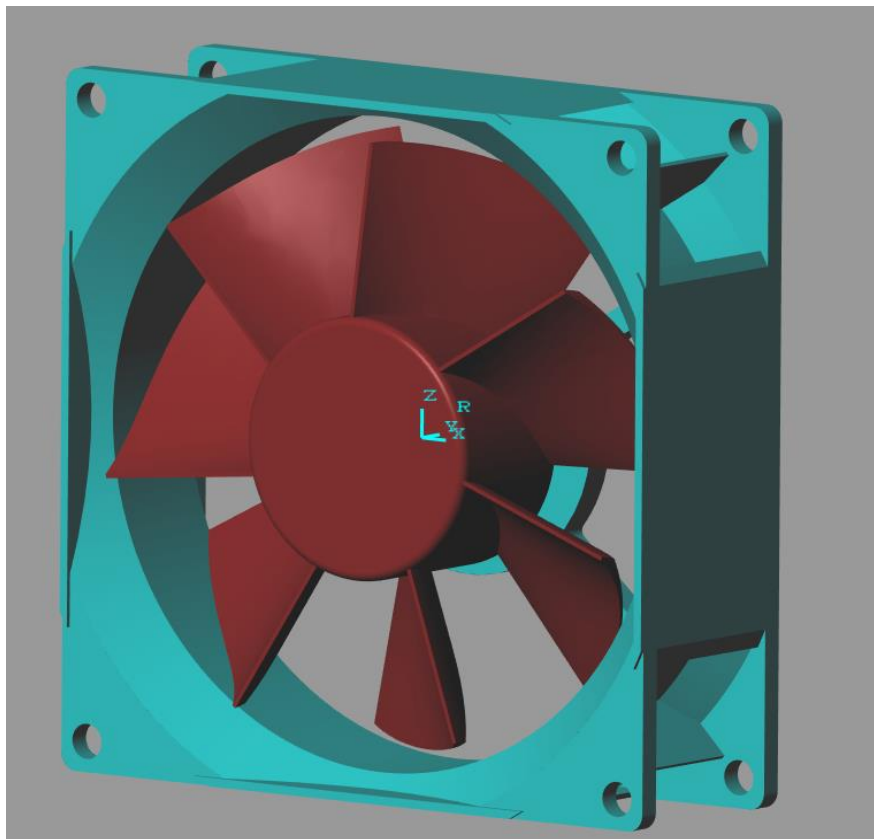


Figure 1

2. INTRODUCTION

Cooling systems are essential in many electronic and mechanical devices to prevent overheating and ensure reliable operation. DC fans are widely used in such cooling systems due to their efficiency and easy control. This project aims to design a DC fan and simulate its operation based on temperature changes. Using SolidWorks, a detailed 3D model of the fan, including blades, shaft, and housing, was created. The design was then imported into MATLAB Simscape, a powerful simulation tool, which allows mechanical systems to be tested virtually. Simulating the fan's behavior helps we understand how it will perform under different temperature conditions without needing a physical prototype. The fan rotates around a revolute joint, driven by torque applied in the simulation. A control logic was added to automatically turn the fan ON when temperature rises and OFF when it cools down. This method shows how mechanical design and control systems can work together to create an effective cooling device. The project demonstrates the value of combining CAD and simulation for product development and testing.

3. METHADODOLOGY

In this project, we designed and simulated a temperature-controlled DC fan system. First, all the mechanical parts of the fan were modeled and assembled in SolidWorks. Then, the model was imported into Simscape to combine the mechanical design with control logic. Finally, the simulation was set up to control the fan's operation based on temperature changes.

3.1 CAD Design in SolidWorks

We created all the fan components such as the propeller blades, fan housing, and shaft using SolidWorks software. These parts were carefully assembled to allow smooth rotation of the fan blades. A revolute joint was used at the connection point to enable realistic rotational movement. Accurate dimensions and proper assembly were important to ensure compatibility for simulation.

3.2 Importing SolidWorks Model into Simscape

Since SolidWorks and Simscape use different file formats, we first exported the fan model as an XML file compatible with Simscape. During the import process, some version compatibility issues were encountered, which we resolved using MATLAB's `smimport` command. This allowed us to successfully bring the model into the Simscape environment, where mechanical and control systems could be integrated.

3.3 Control Logic and Simulation Setup

We developed control logic that switches the fan ON when the temperature rises above a set threshold and OFF when it cools down. The fan's rotation was controlled by applying torque at the revolute joint. To simulate a smooth start-up, a ramp input was used. This setup allowed us to realistically simulate the fan's behavior in response to temperature changes

4. SIMSCAPE MODEL OF THE SYSTEM

This section shows the complete Simscape model. It includes the fixed fan housing and the rotating propeller connected by a revolute joint. The torque input drives the fan blades to rotate. Control signals from the temperature sensor determine when the fan turns ON or OFF. The model integrates mechanical and control parts for a realistic simulation. As shown in figure 2

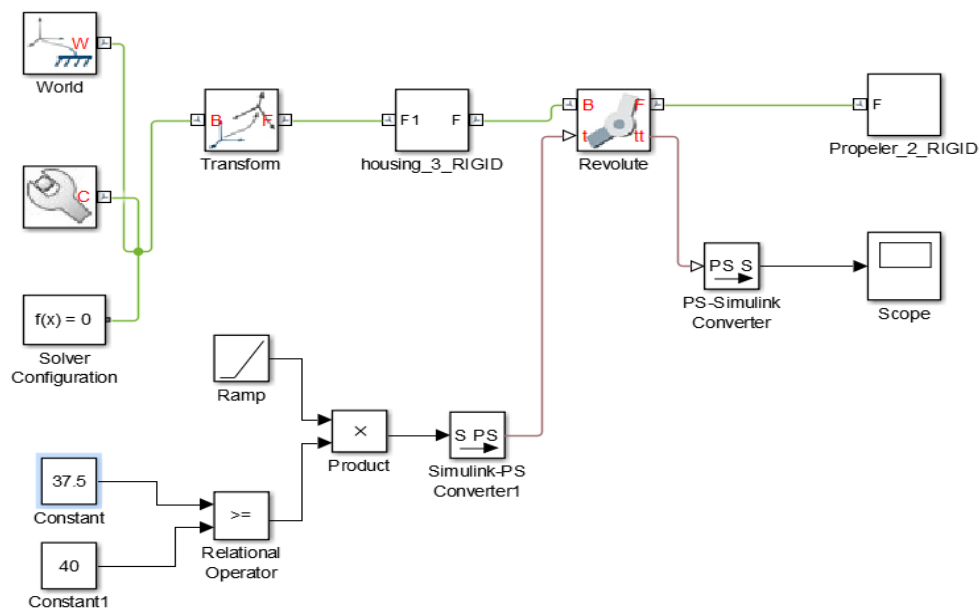


Figure 2

Table: 2A

No.	Block Name	Parameters / Description
1	Constant (Temperature)	Temperature input set to 37.5°C
2	Constant1 (Threshold)	Temperature limit set to 40°C

No.	Block Name	Parameters / Description
3	Relational Operator ($\geq$)	Compares temperature with threshold to decide fan ON/OFF
4	Ramp	Used for smooth fan startup Slope = +2 Initial Value = 0 Start Time = 1 s
5	Product	Combines ramp signal with temperature condition
6	Simulink–PS Converter	Converts Simulink signal into physical torque signal

Theoretical Calculation

Given

Voltage $V=12\text{V}$
Current $I=0.25\text{A}$
Speed $=100\text{RPM}$

Solution:

Power Calculation

$$P=V \times I$$

$$P=12 \times 0.25$$

$$P=3\text{W}$$

RPM $\rightarrow$ rad/s

$$\omega = 2\pi \times \text{RPM} / 60$$

$$\omega = 100 \times 2\pi / 60$$

$$\omega = 10.47 \text{ rad/s}$$

Torque Calculation

$$T = P / \omega$$

$$T = 3 / 10.47$$

$$T = 0.286 \text{ N}$$

$$T \approx 0.29 \text{ N at } 100 \text{ RPM}$$

The rotational speed of 100 RPM was manually set in SolidWorks for the DC fan model. The average operating voltage (12 V) and current (0.25 A) of the DC fan were used to calculate the input power.

Based on these values, the theoretical torque was calculated as approximately 0.29 Nm at 100 RPM under ideal conditions

5. RESULT

The simulation results showed that the fan remained OFF at low temperatures. When the temperature increased, the fan started smoothly and maintained rotation until the temperature dropped again. The control system worked effectively to turn the fan ON and OFF based on temperature changes. The model behaved as expected, confirming that the design and control strategy are correct.

5.1 Scope Analysis of Revolute Joint Torque

This analysis explains the torque applied at the fan's revolute joint. The graph shows three curves: the input torque to start the fan, the reaction torque at the joint, and the load torque from the fan blades. The smooth rise in torque shows how the fan starts without sudden jerks, ensuring stable operation. As shown in Figure 3 and Figure 4.

When Temperature is Low to the Required Limit.



Figure 3

When Temperature is Equal or high to the Required Limit.

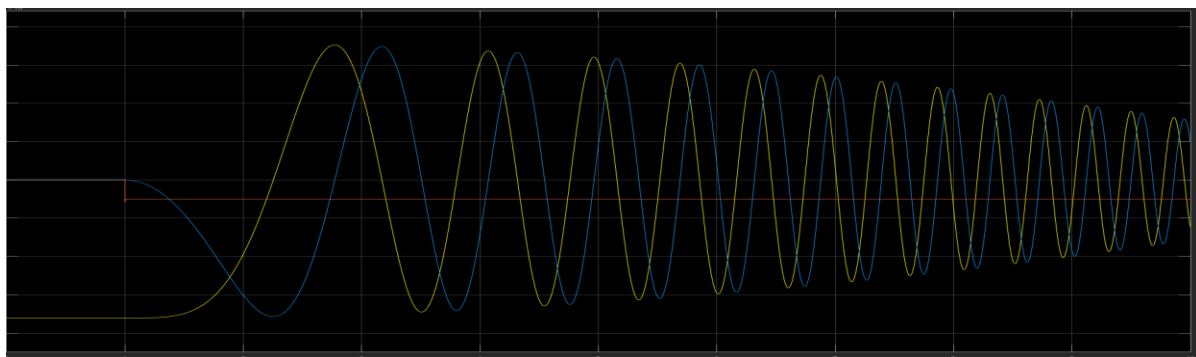


Figure 4

These graph shows the dynamic response of the fan system to a change in temperature set point. When the set point changes, the system exhibits oscillations due to inertia and control action. These oscillations gradually decay, and the system reaches a stable steady-state, indicating proper damping and effective temperature-based control.

By Increasing the Damping coefficient

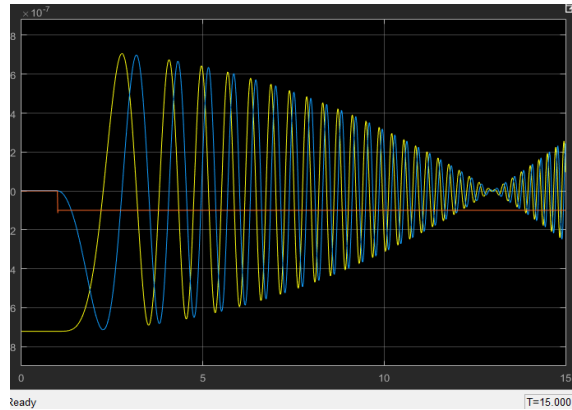


Figure 5

By Decreasing the Damping Coefficient

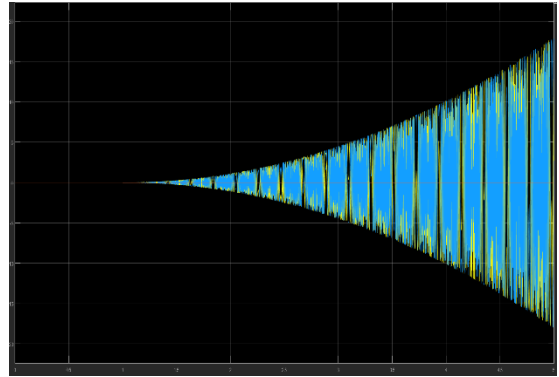


Figure 6

The damping coefficient controls the smoothness of the assembly motion. By adjusting this value, jerking can be reduced as shown in figure 5 and figure 6. In my project we set the coefficient value of damping is 1 to run smoothly without jerking in Figure 4.

6. CONCLUSION

In conclusion, the project was successful in designing a DC fan and simulating its operation with temperature control. We overcame challenges related to software compatibility and created a smooth control logic to turn the fan ON and OFF. The simulation results confirmed that the fan model and control system work together effectively. This project demonstrates how CAD design and simulation can be combined to study real-world mechanical systems before physical implementation.